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Experimental and Theoretical Study on CHF of a Ultra-Supercritical Circulating Fluidized Bed Boiler Water-Wall Tube at near-Critical Pressures

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Abstract: The experimental and theoretical research on the critical heat flux (CHF) in a uniformly heated water-wall tube of the efficient ultra-supercritical circulating fluidized bed (USCFB) boiler has been conducted. In particular, the experimental pressure varies from 18 MPa to 21 MPa, which is from $0.814P_{cr}$–$0.95P_{cr}$ (P_{cr}: critical pressure). The mass flux varies from 310 $kg\cdot m^{-2}\cdot s^{-1}$ to 550 $kg\cdot m^{-2}\cdot s^{-1}$. The inlet sub-cooling temperatures vary from 5°C to 10°C. The material of the tube is 12Cr1MoVg. From experimental investigation, the near critical pressure CHF test data of water are obtained. We find that the CHF mainly occurs when the vapor qualities are less than 0.4, and it occurs earlier (at lower vapor quality) when the pressure is closer to 22.115 MPa or the mass flux is smaller. From the experimental data, a correlation function for the CHF is established via regression and machine learning. Correlations established via machine learning greatly improved the regression accuracy. To study the CHF phenomenon mechanically, a theoretical model is established based on the near-surface bubble crowding model describing the DNB-type CHF. In the development of the CHF model, the friction resistance coefficient is determined according to our test results. By comparison with different experimental results, the near-surface bubble crowding model is well suited to describe DNB-type CHF. The calculation results of the model can provide reference for the optimal design of the USCFB boiler.

Keywords: circulating fluidized bed, critical heat flux, near-critical pressure, bubble crowding, machine learning

1. Introduction

Comparing with the single phase heat transfer, boiling is a more effective heat transfer method with a larger heat transfer coefficient (HTC) and smaller temperature difference, which has been applied in various industrial equipment. Although boiling can facilitate heat transfer, the phenomenon of heat transfer deterioration seems inevitable under some extreme conditions, where increasing temperature appears as the heat flux reaches a threshold. The rapidly increased temperature could damage equipment and even cause serious accidents. The

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Nomenclature

A	cross section are/m^2	Q_E	input electric power/kW
Bo	boiling number	q	heat flux/kW·m^{-2}
C_0	distribution parameter	q_{cr}	critical heat flux/kW·m^{-2}
c_p	specific hear/J·kg^{-1}·K^{-1}	q_{eva}	heat flux required for evaporation in the bubbly layer/kW·m^{-2}
D_b	bubble departure diameter/m	Re	Reynolds number
d	tube diameter/m	U_c	velocity/m·s^{-1}
d_{in}	inner tube diameter/m	T	temperature/°C
E	input voltage/V	T_{in}	inlet fluid temperature/°C
F_d	drag force/N	T_{out}	outlet fluid temperature/°C
f	frictional resistance coefficient	T_{sat}	saturated temperature/°C
G	mass flux/kg·m^{-2}·s^{-1}	T_{wi}	inner tube wall surface temperature/°C
G^*	limited transverse mass flux/kg·m^{-2}·s^{-1}	T_{wo}	outer tube wall interface temperature/°C
g	acceleration of gravity/m·s^{-2}	$\overline{U_b}$	the average flow velocity in the bubble layer/m·s^{-1}
H	specific enthalpy/kJ·kg^{-1}	U_c	the velocity of the main flow at the interface of the two regions/m·s^{-1}
H_{fg}	latent heat/kJ·kg^{-1}	$\overline{U_c}$	the average velocity of the main flow/m·s^{-1}
ΔH	added enthalpy/kJ·kg^{-1}	U^+	dimensionless velocity
ΔH_{in}	inlet fluid subcooled enthalpy/kJ·kg^{-1}	V_{gj}	drift velocity/m·s^{-1}
$\Delta H_{sub,d}$	fluid subcooled enthalpy for the OBD/ kJ·kg^{-1}	x	vaper quality
h	heat transfer coefficient/kW·m^{-2}·K^{-1}	$x_{d,OBD}$	thermodynamic equilibrium for the OBD
h_s	single phase heat transfer coefficient/ kW·m^{-2}·K^{-1}	x_i	inlet vapor quality
I	input current/A	x_e	outlet vapor quality
k	thermal conductivity/W·m^{-1}·K^{-1}	x_{tr}	true flow quality
k_0	thermal conductivity when temperature is equal to 0°C/W·m^{-1}·K^{-1}	Y^+	dimensionless distance
k_w	thermal conductivity of wall metal/ W·m^{-1}·K^{-1}	$Y_B{}^+$	dimensionless distance from wall surface to the top of vapor bubble
L	length of the test section/m	z	axial location/m
N_b	the number of wall-attached bubbles	**Greek symbols**	
N_d	liquid-vaper density ratio	α	void faction
N_{sub}	subcooling number	α_{tr}	average void fraction
p	pressure/MPa	β	volume void fraction
p_r	reduced pressure/MPa	γ	thermal conductivity coefficient varying with temperature
Pr	Prandtl number	δ_b	thickness of bubble layer/m
Δp	pressure drop/MPa	ε	wall roughness/m
Δp_a	acceleration pressure drop/MPa	ξ	perimeter/m
Δp_f	frictional pressure drop/MPa	ξ_i	perimeter of the interface of the wall bubbly layer and core/m
Δp_g	gravitational pressure drop/MPa	ξ_w	the perimeter of the heated tube wall/m
Δp_{lo}	frictional pressure drop in liquid-phase zone/MPa	η	thermal efficiency
Δp_{tp}	frictional pressure drop in two-phase zone/MPa	η_c	fraction of cross-section

(Continued Symbol table)

θ	bubble contact angle/(°)	cr	critical
λ	skin friction coefficient	eq	thermal equilibrium
μ	dynamic viscosity/N·s·m^{-2}	f	fluid
ρ	density/kg·m^{-3}	g	saturated vapor
σ	surface tension coefficient/N·m^{-1}	i	interface between bubbly layer and core
τ_i	viscous shear stress of the core/N·m^{-2}	l	saturated liquid
τ_w	apparent wall shear stress/N·m^{-2}	tr	true
$\tau_{w,v}$	viscous shear stress on wall/N·m^{-2}	w	heated wall
Φ_{acc}	acceleration term	2ϕ	two phase
χ	resistance coefficient of metal	ϕ_{LO}^2	two phase frictional multiplier
χ_0	resistance coefficient of metal when temperature is equal to 0°C	**Abbreviations**	
ω	resistance coefficient varying with temperature	CFB	circulating fluidized bed
Subscript		CHF	critical heat flux
avg	average	DNB	departure from nucleate boiling
b	bubble layer	HTC	heat transfer coefficient
bc	from bubbly	NIST	national institute of standards and technology
c	core	REFPROP	reference fluid thermodynamic and transport properties database
cb	from core to bubble layer		

value of heat flux at the critical point is called CHF, which is also known as boiling crisis and has become an important parameter to characterize boiling heat transfer deterioration.

Flow boiling is widely used in engineering and two typical types of CHF phenomena are observed in this type of boiling. One is the departure from nucleate boiling (DNB) occurring in the subcooled boiling zone or low vapor quality zone. The DNB appears when the bubble formation rate is higher than the bubble detachment rate, which makes the bubble crowd near the wall and the wall lose contact with sufficient liquid. The other is the surface dryout typically observed in the high vapor quality zone. In practical industrial applications, CHF prediction plays a critical role in preventing operation risks and reducing equipment failures.

CFB is an efficient and clean low-quality coal combustion technology. The research direction of CFB technology is to combine itself with ultra-supercritical technology to improve the thermal efficiency [1]. In this work, we concern the 660 MW efficient USCFB boiler [2, 3], which is the first USCFB boiler in the world. For this type of boiler, the design of water wall is the key technology and the CHF is a critical parameter in the design of water wall. Recently, we have studied the heat transfer, resistance and instability characteristics of the water-wall tube [4–6]. However, the CHF phenomenon near critical pressure for the water-wall tube has yet to be understood.

Although the 660 MW efficient USCFB boiler works at supercritical pressure and no phase change at the rated operating conditions, it should be noted that the boiler inevitably needs to be operated at near-critical pressure to start up and shut down or to realize the in-depth peak regulation. In present work, we focus on the CHF phenomenon in the water-wall tube of the 660 MW efficient USCFB boiler near critical pressures.

In literature, the experimental studies on the CHF phenomenon have been concentrated on low pressure or using the fluid modeling method. For example, Kossolapov et al. [7] experimentally revealed the induction mechanism of flow boiling CHF with different subcooling at atmospheric pressure. Liu et al. [8] analyzed the flow regime transition after the occurrence of CHF by calculating Weber number. Liu et al. [9] analyzed the flow boiling CHF in microgravity. According to the above analysis, the CHF experimental data of water near critical region (18–21 MPa) is scarce due to the expensive cost and difficulty of the experiment.

The fluid modeling method replaces water by an organic liquid considering some similarity of their physical properties, which has been widely used in the CHF experiments at low boiling points and low pressures.

Typically, the organic liquid with low pressure, temperature and latent heat, such as R134a, was selected in practical experiments, by which the difficulty in operating with water can be circumvented while ensuring the understanding drawn from the experimental studies is applicable to water [10–13]. However, it is difficult to maintain all similarity conditions satisfied in fluid modeling experiments, especially for complex phenomena such as vapor-liquid two-phase boiling flow [14, 15]. The real-fluid experimental results can provide data to validate the accuracy of results obtained for modeling fluids. Thus, it is necessary to directly study the CHF phenomenon of water in experiments, especially near the critical pressures.

Machine learning is a method in which a computer uses the acquired data to create a model, and then uses that model to make predictions. This method has advantageous for analyzing the CHF phenomena [16, 17]. Therefore, using machine learning method to study CHF phenomenon near critical pressure region is an effective way.

To this end, we built a large water-steam recycling system, which can be operated under the working conditions of the 660 MW efficient USCFB boiler. Using that, we experimentally studied the CHF phenomenon near critical pressures. In particular, we examined the thermal properties of near-critical water and the influence of various parameters on CHF. Through regression and machine learning techniques, we inferred the correlation function between the nondimensional CHF and different parameters from experimental data.

In regard to theoretical studies, researchers have developed a variety of theoretical models and adopted different methods in literature. For instance, Kam et al. [18] developed a CHF prediction model based on considering various effect factors. Zhang et al. [19] modified the interface separation model to study the influence of gravity on CHF. Based on certain hypothesis and simplification about the CHF occurrence mechanism, a theoretical model allows for analytical or numerical prediction of the CHF, which can be much more efficient and less demanding than experimental studies [20–23]. The theoretical models of the CHF commonly employed in literature include the bubble crowding model, the micro-liquid layer evaporation model at the bottom of the bubble layer, the interface separation model, and the liquid film evaporation model. For the DNB-type which occurs in the subcooled zone and saturated boiling zone with low dryness, Weisman and Pei [20] put forward the near-surface bubble crowding model in 1983. In this model, the flow field is formed by bubble layer and mainstream zone. When bubble layer congestion formed near the wall occurs, heat exchange is impeded between the mainstream zone and the heated surface. As the wall temperature rises, the CHF is triggered.

In this work, the CHF was estimated from the abnormal increase of wall temperature (rapidly exceeded 620°C). CHF mainly occurs in cross section 9 and all vapor qualities of abnormal wall temperature increase were less than 0.4. Therefore, the CHF in our test was regarded as the DNB-type CHF caused by bubble crowding near the wall surface. We adopted the near-surface bubble crowding model to establish a CHF calculation model. The established model is suitable for the low-dryness region of the smooth tube near critical pressures. We validated the theoretical model by the test data from different researchers. The model's predictions can provide reference for the optimal design of the USCFB boiler.

2. Experimental Method and Data Processing

2.1 Experimental system and procedures

The test system is described as Fig. 1. During a test, the high-pressure plunger pump drove the deionized water into two routes: main loop and bypass. In both the preheating and test sections of the experimental system, an alternating current was used to convert the electric energy into heat energy through the thermal effect of electronic currents. The fluid flow through the loop was directly heated by the resistance of the tube itself [24, 25]. Fig. 2 shows the actual experimental loop used in this work.

During a test, the pressure and mass flux were fixed. After the inlet subcooling was fixed at 5–10°C, the power of the test section was gradually increased by 1–2 kW each time. After each increase of the test section power, 30 sets of data were recorded once the system achieved a steady state. The value of CHF was then estimated by the abnormal increase of wall temperature (rapidly exceeded 620°C).

2.2 Test section

The test section was a tube of 2 m long made from the material of 12Cr1MoVg. The diameter and thickness of the test section are 30 mm and 5.5 mm, respectively. The fluid temperatures were measured via armored thermocouples. The outer wall temperatures of nine cross sections on the test tube were measured via K-type thermocouples welded on each cross section. The pressure and differential pressure were measured using the Rosemount pressure and differential pressure transmitter. Mass flux was measured using a RHEONIK mass flow meter. Fig. 3 shows the diagram of test section. All measured signals in the experiment were converted to electrical signals and transmitted to an industrial personal

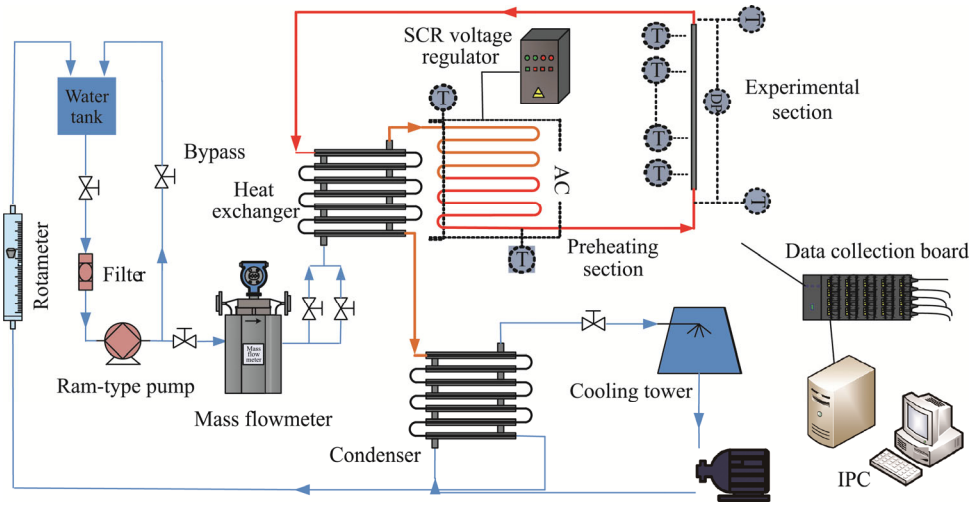

Fig. 1 Diagram of the test system

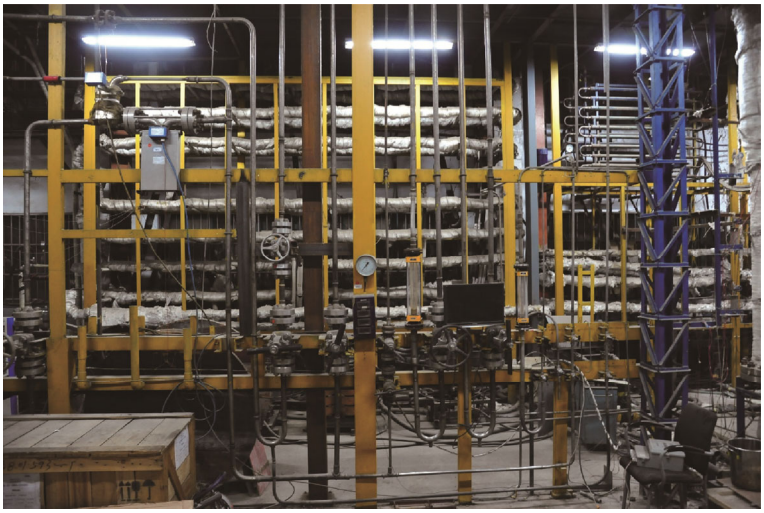

Fig. 2 Actual picture of the test system

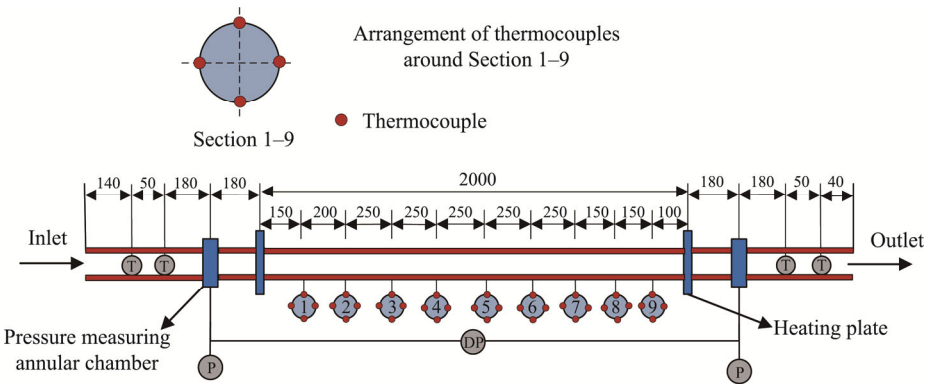

Fig. 3 Diagram of the test section (unit: mm)

computer by the IMP3595 data acquisition board. The coefficients of thermal conductivity and electrical resistance were calculated as Eqs. (1) and (2), respectively:

$$k_w=k_0(1+\gamma T) \quad (1)$$

$$\chi=\chi_0(1+\omega T) \quad (2)$$

where the coefficients of thermal conductivity and electrical resistance are presented in Table 1.

Table 1 Parameter values for the thermal conductivity and electrical resistance of 12Cr1MoVg

Material	ω/°C^{-1}	γ/°C^{-1}	χ_0/Ω·m	k_0/ kW·m^{-1}·s^{-1}
12Cr1MoVg	0.0102	-0.455×10^{-3}	0.110×10^{-6}	0.0494

2.3 Data processing

The η of the test section is given as follows:

$$\eta = \Delta H / Q_{\mathrm{E}} \tag{3}$$

where ΔH and Q_{E} are calculated from:

$$\Delta H = \frac{1}{4}\pi d_{\mathrm{in}}^2 G c_p \left(T_{\mathrm{out}} - T_{\mathrm{in}}\right) \tag{4}$$

$$Q_{\mathrm{E}} = EI \tag{5}$$

The q is given by:

$$q = \frac{Q_{\mathrm{E}}\eta}{\pi d_{\mathrm{in}} L} \tag{6}$$

The HTC is defined as:

$$h = \frac{q}{T_{\mathrm{wi}} - T_{\mathrm{f}}} \tag{7}$$

where T_{wi} is determined as:

$$T_{\mathrm{wi}} = T_{\mathrm{wo}} - \frac{q d_{\mathrm{in}}}{2k_{\mathrm{w}}}\left[\frac{1}{2} - \frac{d_{\mathrm{in}}^2}{d_{\mathrm{out}}^2 - d_{\mathrm{in}}^2}\ln\left(\frac{d_{\mathrm{out}}}{d_{\mathrm{in}}}\right)\right] \tag{8}$$

The Δp_{f} can be calculated from:

$$\Delta p_{\mathrm{f}} = \Delta p - \rho g L \tag{9}$$

Further, the two phase frictional pressure drop $\Delta p_{\mathrm{f,2\phi}}$ is given by:

$$\Delta p_{\mathrm{f,2\phi}} = \Delta p - \Delta p_{\mathrm{g}} - \Delta p_{\mathrm{a}} \tag{10}$$

where Δp_{g} and Δp_{a} can be calculated from:

$$\Delta p_{\mathrm{g}} = \frac{\rho_{\mathrm{l}} g L}{\left(x_{\mathrm{e}} - x_{\mathrm{i}}\right)\left(\rho_{\mathrm{l}}/\rho_{\mathrm{g}} - 1\right)}\ln\left[\frac{1 + x_{\mathrm{e}}\left(\rho_{\mathrm{l}}/\rho_{\mathrm{g}} - 1\right)}{1 + x_{\mathrm{i}}\left(\rho_{\mathrm{l}}/\rho_{\mathrm{g}} - 1\right)}\right] \tag{11}$$

$$\Delta p_{\mathrm{a}} = G^2\left(x_{\mathrm{e}} - x_{\mathrm{i}}\right)\left(\frac{1}{\rho_{\mathrm{g}}} - \frac{1}{\rho_{\mathrm{l}}}\right) \tag{12}$$

The fluid enthalpy H of each cross section can be calculated from:

$$H_{\mathrm{f}} = H_{\mathrm{in}} + \frac{4q\Delta l}{Gd} \tag{13}$$

The vapor quality of fluid x is defined as:

$$x = \frac{H_{\mathrm{f}} - H_{\mathrm{l}}}{H_{\mathrm{fg}}} \tag{14}$$

2.4 Uncertainty analysis

Table 2 summarized the uncertainties [26]. More specifically, for the directly measured parameter X_n, the parameter and its uncertainty (δX_n) can be expressed as:

$$X_n = X_{n,\mathrm{m}} \pm \delta X_n \tag{15}$$

where δX_n represents the maximum permissible error of measuring instrument; $X_{n,\mathrm{m}}$ is the measured value of X_n.

For the calculated parameter R, we assume

$$R = R(X_1, X_2, X_3, \cdots, X_m) \tag{16}$$

where X_n represents an independent, measured parameter. The uncertainty of R can then be calculated as:

$$\delta R = \sqrt{\sum_{n=1}^{m}\left(\frac{\partial R}{\partial X_n}\delta X_n\right)^2} \tag{17}$$

where δR is contributed by the uncertainty of each X_n.

The maximum relative uncertainty of a measured or calculated parameter was finally obtained by dividing the uncertainty by the parameter's minimum value.

Table 2 Uncertainties analysis

Parameter	Type	Unit	Maximum relative uncertainty
Pressure	Measured	MPa	0.55%
Pressure difference	Measured	kPa	3.76%
Fluid temperature	Measured	°C	0.29%
Wall temperature	Measured	°C	0.32%
Mass flux	Measured	kg·m^{-2}·s^{-1}	1.44%
Heat flux	Calculated	kW·m^{-2}	5.73%
HTC	Calculated	kW·m^{-2}·K^{-1}	11.42%

3. Experimental Results and Analysis

We first summarize and analyze the thermal properties of near-critical pressures water and the effects of different parameters on the CHF, including the pressure and mass flux. In this work, the CHF was estimated from the abnormal increase of wall temperature (rapidly exceeded 620°C). CHF mainly occurs in cross section 9. All vapor qualities of abnormal wall temperature increase were less than 0.4. Therefore, the CHF in our test was regarded as the DNB-type CHF. From the experimental data, the correlation functions for the CHF were also inferred via regression and machine learning techniques.

3.1 Thermal properties of near-critical water

The thermal properties of water, including density, viscosity, thermal conductivity and specific heat change, can vary with pressure or temperature. Thus, we first plotted the thermal properties of water under different pressures in Fig. 4, where saturated temperature, latent heat of vaporization and the ratio of saturated steam to saturated water are used as the y-axis to express the variation trend of physical parameters. Fig. 4 shows that the thermal properties of water fluctuate drastically with pressure or temperature in the regime of 18–22 MPa. The data for the properties of water was obtained from the NIST REFPROP.

Next, we examined the variation of thermal properties with the pressure fixed at 21 MPa, as depicted in Fig. 5.

We find that in the region near the saturated temperature, the density, viscosity, thermal conductivity and specific heat change dramatically with the temperature.

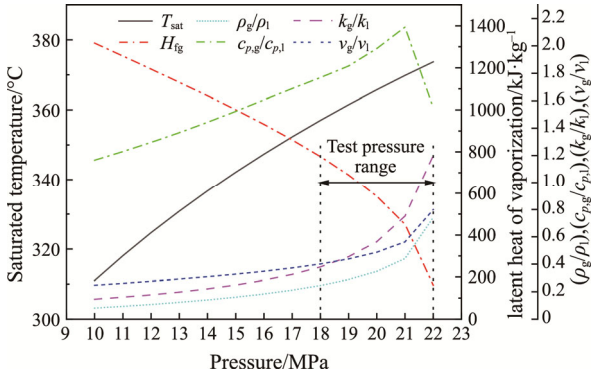

Fig. 4 Thermal properties of water near critical pressures

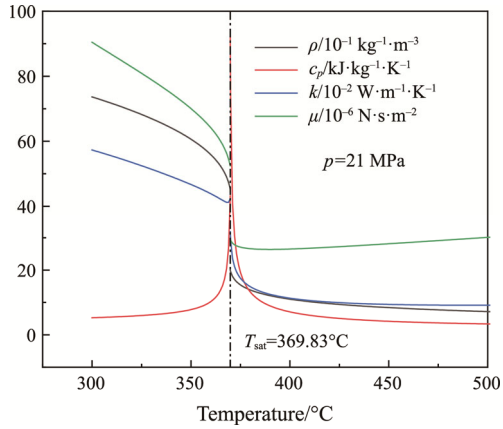

Fig. 5 Thermal properties of water vs. temperature at a fixed pressure 21 MPa

The thermal properties variations of water near critical pressures indicate that the CHF phenomenon in this region should be investigated.

3.2 Effect of pressure on CHF

Because the thermal properties of water vary drastically with the increase of pressure at near-critical pressures, it is essential to investigate the influence of pressure on CHF. Thus, we examined the T_{wi} and HTC to detect the heat transfer deterioration. In particular, the T_{wi} and HTC were calculated with increasing bulk enthalpy. During the test, we realized the increase of enthalpy by increasing the heat flux. In Fig. 6, four different near-critical pressures were considered from 18 MPa to 21 MPa when the mass flux was fixed at 440 kg·m^{-2}·s^{-1}. Fig. 6(a) depicts the enthalpy values of wall temperature rise as 2010.9, 1982.7, 1938.5 and 1903.2 kJ·kg^{-1} at four near-critical pressures, respectively, whose corresponding vapor qualities are 0.358, 0.298, 0.19 and 0.034. The density difference between saturated steam and water decreases when the pressure is closer to 22.115 MPa, which reduces the buoyancy and makes it more difficult for bubbles to escape from the wall. As a result, the CHF decreases, and heat transfer deteriorations occur earlier. From the results in Fig. 6(b), we can see the HTC increases when the boiling heat transfer occurs and the HTC decreases rapidly when heat flux reaches the CHF. The HTC decreases at lower enthalpy when the pressure increases.

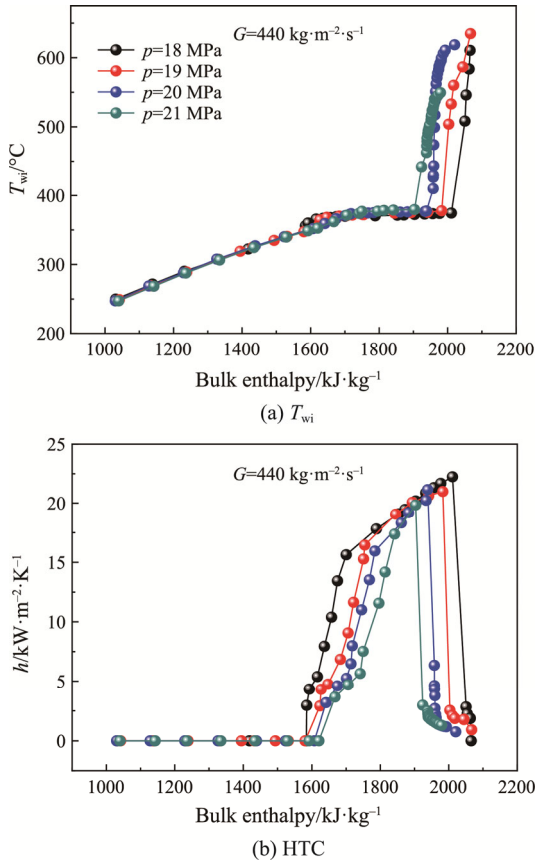

Fig. 6 Effect of pressure

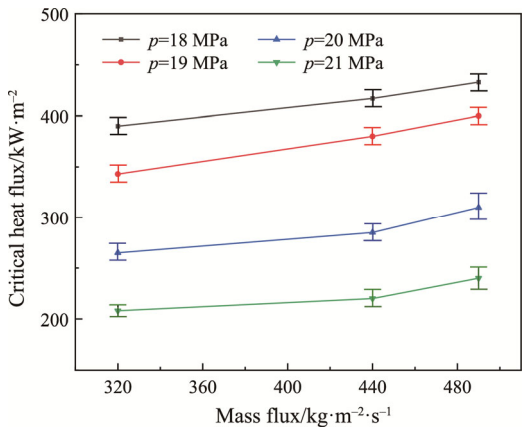

Fig. 7 Effect of pressure and mass flux on CHF

Fig. 7 shows the value of CHF at different pressures. The CHF decreases when the pressure is close to 22.115 MPa. When mass flux is equal to 440 kg·m^{-2}·s^{-1}, the corresponding CHF are found to be 417, 380, 285, and 220 kW·m^{-2}, respectively, at four different near-critical pressures from 18 MPa to 21 MPa.

3.3 Effect of mass flux on CHF

Here, we fixed the pressure at 20 MPa and calculated the T_{wi} and HTC of water when the mass fluxes are equal to 320, 440, and 490 kg·m^{-2}·s^{-1}. The T_{wi} rises later for larger mass flux. It is because a larger mass flux can take away bubbles near the wall, and as a result, more working medium can exchange heat with the wall. From the calculations on the T_{wi} and HTC, as shown in Fig. 8(a), we can see the bulk enthalpy reduced from 1980.9 to 1938.5 to 1911.5 kJ·kg^{-1} and the vapor qualities decreasing from 0.263 to 0.19 to 0.144 as the mass fluxes decrease from 490 to 440 to 320 kg·m^{-2}·s^{-1}. Fig. 8(b) shows that the HTC increases when the boiling heat transfer occurs and the HTC decreases rapidly when heat flux reaches the CHF. The HTC decreases at lower enthalpy when the mass flux is smaller.

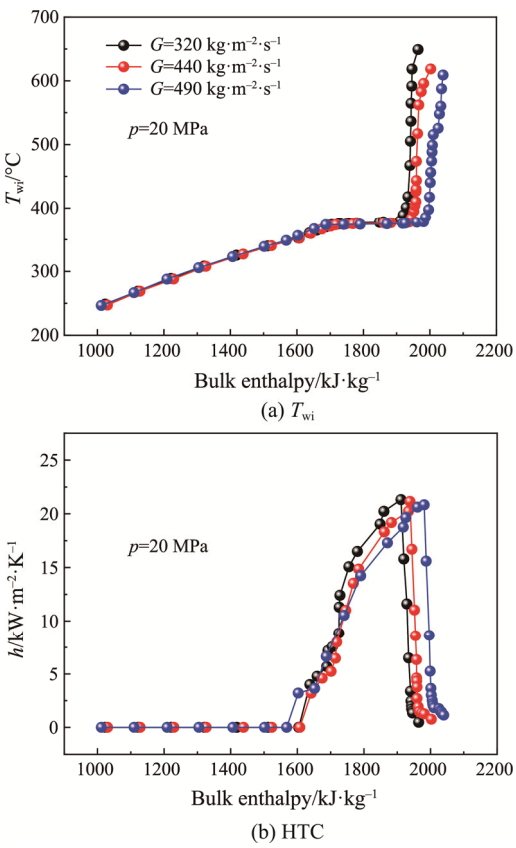

Fig. 8 Effect of mass flux

Fig. 7 depicts that the values of the CHF are equal to be 265, 285 and 310 kW·m^{-2}, respectively, increasing with the mass flux. Here, the pressure is fixed at 20 MPa, and the mass fluxes are: 320, 440, and 490 kg·m^{-2}·s^{-1}, respectively.

3.4 Correlation

When the tube diameter and heated length are constant, the pressure, mass flux and inlet subcooling are the key parameters determining the CHF. Thus, we next sought a function to relate the CHF to those parameters, i.e.,

$$q_{cr} = f_1\left(p, G, \Delta H_{in}\right) \tag{18}$$

Using dimensional analysis, the CHF, pressure, mass flux, and inlet subcooling are nondimensionalized by characteristic thermodynamic parameters under the CHF conditions, leading to the following dimensionless numbers:

$$Bo = q_{cr}/GH_{fg} \tag{19}$$

$$p_r = p/p_{cr} \tag{20}$$

$$Re = Gd/\mu \tag{21}$$

$$N_{sub} = \Delta H_{in}/H_{fg} \tag{22}$$

Thus, Eq. (18) can be written in a dimensionless form:

$$Bo = f_2\left(p_r, Re, N_{sub}\right) \tag{23}$$

With

$$N_d = \rho_l/\rho_g \tag{24}$$

Eq. (23) can be further rewritten as [10]:

$$Bo = f_3\left(N_d, Re, N_{sub}\right) \tag{25}$$

The functional form of f_3 in Eq. (25) was then inferred from our experimental data for Bo>0.002. In particular, we employed three methods to determine the functional form of f_3 from the training data, including the multiple linear regression [27], multivariate Gaussian Process regression (GPR) [28], and deep neural network (DNN) [29, 30]. The performance of three methods was compared. For the sake of cross validation and minimizing the bias of data, 520 experimental data were randomly divided into five groups. The data for training and test were combined from the five groups by permutation, which led to five combinations. In each combination, four groups were used as the training data for regression, and the fifth group was used as the test data for validation. Each such combination led to a relative error of regression defined as:

$$\varepsilon_r = \frac{\left\|Bo'_p - Bo'\right\|_{L_2}}{\left\|Bo'\right\|_{L_2}} \times 100\% \tag{26}$$

where $Bo'_p = \left[Bo'_{p,1}, Bo'_{p,2}, \cdots, Bo'_{p,m}\right]^T$ is calculated by the output predicted by the function trained from the training data given the test input, and $Bo = \left[Bo'_1, Bo'_2, \cdots, Bo'_m\right]^T$ is the test data. The average ε_r of all five combinations was defined as the mean relative error.

3.4.1 Multiple linear regression

A specific functional form of f_3 in Eq. (25) was assumed [10]:

$$Bo = 10^{C_0} N_{\mathrm{d}}^{C_1} Re^{C_2} N_{\mathrm{sub}}^{C_3}, \tag{27}$$

where C_0, C_1, C_2, and C_3 are the coefficients to be determined. Taking the logarithm to the base 10 of Eq. (27), we obtained a linear functional:

$$y = C_0 + C_1 x_1 + C_2 x_2 + C_3 x_3 \tag{28}$$

where y=lg Bo, x_1=lg N_{d}, x_2=lg Re and x_3=lg N_{sub}. In this paper, we used the multiple linear regression implemented in Origin [31] to determine C_0, C_1, C_2, and C_3 in Eq. (28).

The multiple linear regression resulted in the mean relative error of 29.84%. And the training with the least relative error of 29.39% determined the values of C_0, C_1, C_2, and C_3 as in the following equation:

$$y = 2.182 - 2.310x_1 - 0.989x_2 - 0.0195x_3 \tag{29}$$

from which we obtained the correlation function as:

$$Bo = 10^{2.182} N_{\mathrm{d}}^{-2.340} Re^{-0.989} N_{\mathrm{sub}}^{-0.0195} \tag{30}$$

The multiple linear regression method can provide a simple functional form; the large regression error may not be satisfactory. Thus, we next explored two machine learning techniques to greatly improve the regression accuracy.

3.4.2 Multivariate Gaussian process regression

In the multivariate GPR method, the function $y = f(x_1, x_2, x_3)$ is considered drawn from a multivariate Gaussian process with the mean function μ and covariance kernel k; i.e., $f \sim GP(\mu, k)$. Defining the input $x=[x_1, x_2, x_3]^{\mathrm{T}}$, the training data set $\{(x_i,y_i)|\ i=1,2,\ldots,n\}$ satisfies:

$$y = \left[y_1, y_2, \cdots, y_n\right]^{\mathrm{T}} \sim N\left(\mu, K_{XX}\right) \tag{31}$$

where $\mu=[\mu_1, \mu_2,\ldots, \mu_n]^{\mathrm{T}}=[\mu(x_1), \mu(x_2),\ldots, \mu(x_n)]^{\mathrm{T}}$, and the covariance matrix K_{XX} is defined as $(K_{XX})_{i,j} = k(x_i, x_j)$. Given the test input $X'=[x'_1, x'_2,\ldots, x'_m]$, the predicted output y'_{p} satisfies:

$$\begin{bmatrix} y \\ y'_{\mathrm{p}} \end{bmatrix} \sim N\left(\begin{bmatrix} \mu \\ \mu' \end{bmatrix}, \begin{bmatrix} K_{XX} + \sigma^2 I & K_{X'X} \\ K_{X'X} & K_{X'X'} \end{bmatrix}\right) \tag{32}$$

where σ^2 is the variance of independent normally distributed noise. The posterior distribution is calculated from:

$$p\left(y'_{\mathrm{p}} | y\right) = N\left(\hat{\mu}, \hat{K}\right) \tag{33}$$

where

$$\hat{\mu} = K_{X'X}\left[K_{XX} + \sigma^2 I\right]^{-1}\left[y - \mu\right] + \mu' \tag{34}$$

$$\hat{K} = K_{X'X'} - K_{X'X}\left[K_{XX} + \sigma^2 I\right]^{-1} K_{XX'} \tag{35}$$

In this paper, we chose Eq. (27) as the mean function:

$$\mu\left(x\right) = C_0 + C_1 x_1 + C_2 x_2 + C_3 x_3, \tag{36}$$

and the squared exponential function as the covariance kernel:

$$k\left(x, x'\right) = \gamma^2 \exp\left(-\frac{1}{2}\frac{\left\|x - x'\right\|_{L_2}^2}{\omega^2}\right) \tag{37}$$

where $C=[C_0, C_1, C_2, C_3]^{\mathrm{T}}$; γ and ω are the hyper-parameters. The following negative log marginal likelihood can be minimized to determine γ, ω and σ^2, [28]:

$$\begin{aligned} -\lg p\left(y | C, \gamma, \omega, \sigma^2\right) = & \frac{1}{2} y^{\mathrm{T}}\left[K_{\mathrm{XX}} + \sigma^2 I\right]^{-1} y \\ & + \frac{1}{2}\lg \mathrm{Det}\left(K_{\mathrm{XX}} + \sigma^2 I\right) \\ & + \frac{N}{2}\lg\left(2\pi\right) \end{aligned} \tag{38}$$

by employing the Quasi-Newton optimizer L-BFGS [28].

Once the hyper-parameters and σ^2 are determined, the output y'_{p} can be predicted from Eq. (34) for any given input x'_{p}, where the mean and kernel functions can be computed by Eqs. (36) and (37), respectively. Besides, the standard deviation of the output, i.e., the uncertainty level of y'_{p}, can be estimated by Eq. (35).

In this work, we trained the multivariate GPR models from five different combinations of training and test data and obtained the mean relative error of only 13.87%. In the multivariate GPR model with the least relative error of 11.39%, $C=[C_0, C_1, C_2, C_3]^{\mathrm{T}}=[3.371, -2.984, -1.251, -0.122]^{\mathrm{T}}$, γ=0.139, ω=0.0205, and σ=0.0477. Given a set of x_1 =lg N_{d}, x_2 =lg Re and x_3 =lg N_{sub}, the mean and uncertainty of y=lg Bo can then be predicted from Eqs. (34) and (35).

3.4.3 Deep Neural Network

The regression by the above two methods assumed a particular form of f_3 as in Eq. (27). Without any prior knowledge of the functional form of f_3 and considering f_3 is generally a complex nonlinear function, deep learning can be a powerful regression method to infer f_3 in Eq. (25) from experimental data. Thus, we also trained a feed forward DNN model to represent f_3, where the neural network has multiple hidden layers and hence can model the complex non-linear correlation between the parameters Bo, N_{d}, Re, and N_{sub}. The theoretical and methodological descriptions about DNN can be found in Refs. [29, 30].

The architecture of the DNN adopted herein is depicted in Fig. 9. Here, N_{d}, Re, and N_{sub} are the inputs, and Bo is the output. The DNN trained from our experimental data has 3 hidden layers, each of which has 128, 64, or 16 nodes, respectively. The loss function used in the training is the mean squared error function, and the Rectified Linear Units is the activation function between layers. The batch size and epochs were chosen as 64 and

5000, respectively. And the Adam optimizer with an initial uniform learning rate 10^{-4} was used to minimize the loss function and to obtain the hyper-parameters in the DNN. All the work herein was conducted using the package TensorFlow. The mean relative error of the trained DNN models is 15.97%. The values of network parameters for the one with the least relative error of 13.58% are included in a file provided in the Supporting Information. Given a set of N_d, Re, and N_{sub}, one can call the file in TensorFlow to predict the value of Bo.

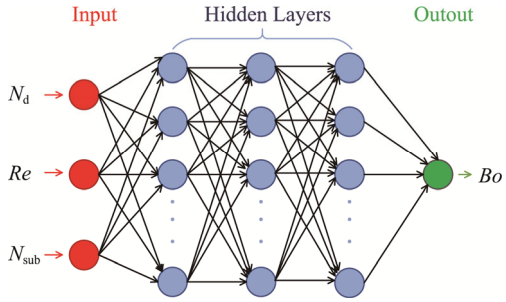

Fig. 9 Architecture of the feed forward DNN adopted in this work

3.4.4 Analysis and comparison

The three regression methods discussed above provided three regression functions (models) that we could use to predict Bo given N_d, Re, and N_{sub}. The application range of above methods is consistent with the experimental parameter range shown in the abstract.

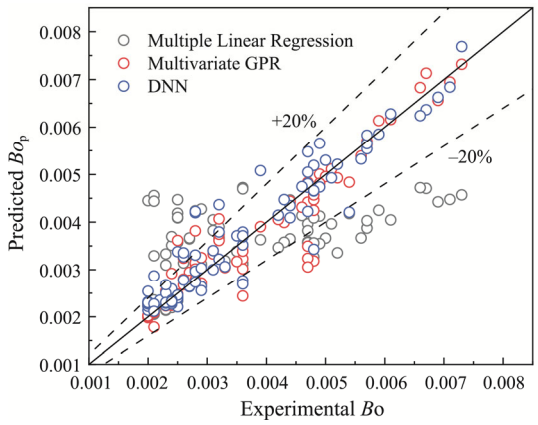

Fig. 10 Comparison of the predicted Bo and experimental Bo

Fig. 10 summarizes the comparison of Bo predicted by the three regression methods with the experimental Bo. We can see from Fig. 10, the machine learning methods such as the multivariate GPR and DNN can provide more accurate regression than the multiple linear regression. In regard to the computational cost of machine learning, the training cost for the GPR is $O(N^3)$, while the inference cost of the GPR scales with $O(N^2)$. For the DNN, the training cost is linear with the sample size N, and the inference cost scales with O(1), which implies the cost of inference is independent of sample size. Thus, if the computational cost is a concern, deep learning via DNN is a better choice than the GPR.

From our experimental studies, we find that the variation of pressure or temperature can significantly alter the thermal properties of the fluid (water) near critical pressure region (18–22 MPa). When the pressure is closer to critical or the mass flux decreases, heat transfer deterioration occurs more easily. The observed CHF is mainly the DNB-type.

4. Theoretical Study and CHF Model

To understand the CHF phenomenon mechanically, we further established an improved DNB-type CHF theoretical model. Since the CHF phenomenon in our tests mainly occurred in the low vapor quality region, the bubble crowding model used to describe the DNB-type CHF was chosen as the basis of our development of the CHF theoretical model.

4.1 Mathematical equations and model derivation

4.1.1 Basic assumptions and physical modeling

We first outline the basic assumptions made to establish the model, which include:

(1) The flow is stable;

(2) A homogeneous flow model can be adopted;

(3) Compared with the system pressure, the variation of axial pressure can be neglected;

(4) The CHF phenomenon occurs when the vapor quality is less than 0.5;

(5) The size of the bubble zone can be ignored compared to the size of the tube.

Fig. 11 shows a structure diagram of the near-surface bubble crowding model. In this model, once the void fraction of the bubble region exceeds a certain value, the energy cannot be carried into the main flow, and the CHF

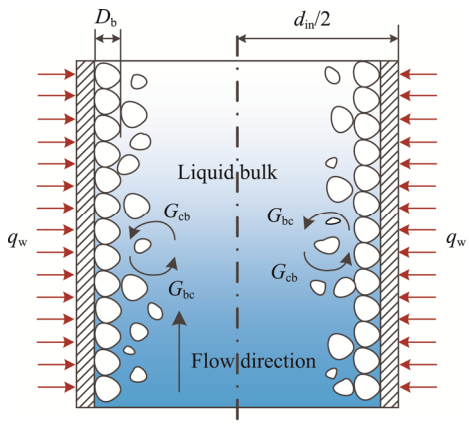

Fig. 11 Structure of the near-surface bubble crowding model, reprinted from Ref. [32]

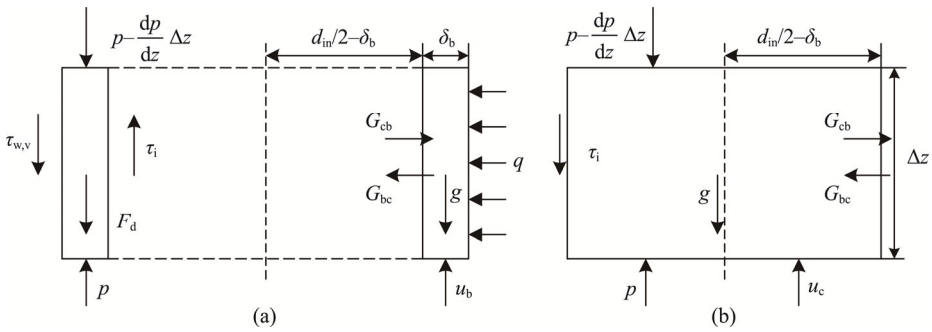

Fig. 12 Control volumes (a) bubble region (b) main-flow region, reprinted from Ref. [32]

phenomenon occurs. Fig. 12 sketches the control volumes of the bubble region and main flow region along with the forces exerted on the control volumes.

4.1.2 Equations of conservation laws

For the bubble region, the conservations of the total and liquid mass lead to the following equations:

$$\frac{dG_b}{dz}=\left(G_{cb}-G_{bc}\right)\frac{\xi_i}{A_b} \tag{39}$$

$$\frac{d\left[G_b(1-x_b)\right]}{dz}=-\frac{q_{eva}\xi_w}{A_bH_{fg}}-\frac{\xi_i}{A_b}\times\left[G_{bc}\left(1-x_b\right)-G_{cb}\left(1-x_c\right)\right] \tag{40}$$

The radial mass transport of the bubble region and main-flow region is constrained due to the stagnant bubble layer, and hence the following identity holds:

$$G_{cb}=G_{bc}=G^* \tag{41}$$

Eqs. (39) and (40) can be combined to obtain the heat flux of evaporation in the bubble layer, as:

$$q_{eva}=\frac{\xi_iH_{fg}G^*\left(x_b-x_c\right)}{\xi_w} \tag{42}$$

The energy conservation equation is given by:

$$q_w\xi_w=\frac{dG_bA_bH_b}{dz}+\xi_i\left(G_{bc}H_b-G_{cb}H_c\right) \tag{43}$$

Combining Eqs. (39) and (43), Eq. (44) can be given:

$$q_w=G_{cb}\left(H_b-H_c\right)\frac{\xi_i}{\xi_w} \tag{44}$$

The pressure, shear force and drag force in the bubble region and main-flow region can be evaluated from the equations of momentum conservation as:

$$-\frac{dp}{dz}-\rho_bg+\frac{\tau_i\xi_i}{A_b}-\frac{\beta'\tau_{w,v}\xi_w}{A_b}-\frac{F_dN_b}{A_bdz}+\frac{\left(G_{cb}U_c-G_{bc}U_b\right)\xi_i}{A_b}=\frac{1}{A_b}\frac{d}{dz}\left(\rho_bU_b^2A_b\right) \tag{45}$$

$$-\frac{dp}{dz}-\rho_cg+\frac{\left(G_{bc}U_b-G_{cb}U_c\right)\xi_i}{A_c}+\frac{\tau_i\xi_i}{A_c}=\frac{1}{A_c}\frac{d}{dz}\left(\rho_cU_c^2A_c\right) \tag{46}$$

Combining Eqs. (45) and (46), the radial mass transport between the bubble region and main-flow region can be obtained as:

$$G^*=\left[-\left(\rho_c-\rho_b\right)g+\frac{\pi d_{in}F_d}{\delta_b^2\left(1-\eta_c\right)A}\right]\frac{A\eta_c\left(1-\eta_c\right)}{\left(U_c-U_b\right)\xi_i} \tag{47}$$

where

$$\eta_c=\frac{\left(d_{in}-2\delta_b\right)^2}{d_{in}^2} \tag{48}$$

Combining Eqs. (44) and (47), the formula used to determine the CHF can be given as [32]:

$$q_{cr}=\left[-\left(\rho_c-\rho_b\right)g+\frac{\pi d_{in}F_d}{\delta_b^2\left(1-\eta_c\right)A}\right]\times\frac{\left(H_b-H_c\right)A\eta_c\left(1-\eta_c\right)}{\left(U_c-U_b\right)\xi_w} \tag{49}$$

The parameters in Eq. (49) can be determined from the constitutive relations discussed in the next section.

4.2 Constitutive relations

4.2.1 The properties of two-phase region

The constitutive relations for the properties of two-phase region are given:

$$\rho_b=\rho_l\left(1-\alpha_b\right)+\rho_g\alpha_b \tag{50}$$

$$\rho_c=\rho_f\left(1-\alpha_c\right)+\rho_g\alpha_c \tag{51}$$

$$H_b=H_l\left(1-x_b\right)+H_gx_b \tag{52}$$

$$H_c=H_f\left(1-x_c\right)+H_gx_c \tag{53}$$

$$x_b=\frac{\rho_g\alpha_b}{\rho_l-\left(\rho_l-\rho_g\right)\alpha_b} \tag{54}$$

$$x_c=\frac{\rho_g\alpha_c}{\rho_f-\left(\rho_f-\rho_g\right)\alpha_c} \tag{55}$$

4.2.2 Void fraction and thickness of bubble region

The α_c [33] is given by:

$$\alpha_b=0.83-0.29\exp\left(-4.71x_{eq}-1.89\right) \tag{56}$$

$$\alpha_c = \frac{1}{\eta_c}\alpha_{tr} - \frac{1-\eta_c}{\eta_c}\alpha_b \tag{57}$$

The average void fraction can be calculated by the model of Zuber and Findlay [34], i.e.,

$$\alpha_{tr} = \frac{x_{tr}}{\left\{C_0\left[x_{tr} + \frac{\rho_g}{\rho_l}\left(1-x_{tr}\right)\right]\right\} + \frac{\rho_g V_{gj}}{G}} \tag{58}$$

where [35]

$$C_0 = \beta\left[1+\left(\frac{1}{\beta}-1\right)^b\right] \tag{59}$$

$$V_{gj} = 1.41\left[\frac{\sigma g\left(\rho_l-\rho_g\right)}{\rho_l^2}\right]^{0.25} \tag{60}$$

$$b = \left(\rho_g/\rho_l\right)^{0.1} \tag{61}$$

$$\beta = \frac{x}{x+\left(1-x\right)\rho_g/\rho_l} \tag{62}$$

4.2.3 True equilibrium quality

The true flow quality distribution can be determined from [36], which is expressed as:

$$x_{tr} = \frac{x_{eq} - x_{d,OBD}\exp\left(x_{eq}/x_{d,OBD}-1\right)}{1-x_{d,OBD}\exp\left(x_{eq}/x_{d,OBD}-1\right)} \tag{63}$$

The equilibrium quality can then be calculated from:

$$x_{d,OBD} = -\Delta H_{sub,d}/H_{fg} \tag{64}$$

4.2.4 Point of bubble departure and departure diameter

The point of bubble departure is also a critical parameter to calculate the CHF, which can be evaluated from [37]:

$$\Delta H_{sub,d} = q\left[\frac{C_{pf}}{h_s} - \frac{Pr_f Y_B^+}{G\sqrt{f/8}}\right] \quad \text{for} \quad Y_B^+ \le 5 \tag{65}$$

$$\Delta H_{sub,d} = q\left\{\frac{C_{pf}}{h_s} - \frac{5}{G\sqrt{f/8}}\left[Pr_f + \ln\left(1+\frac{Pr_f Y_B^+}{5} - Pr_f\right)\right]\right\} \quad \text{for} \quad 5 < Y_B^+ \le 30 \tag{66}$$

$$\Delta H_{sub,d} = q\left\{\frac{C_{pf}}{h_s} - \frac{5}{G\sqrt{f/8}}\left[Pr_f + \ln\left(1+5Pr_f\right) + \frac{1}{2}\ln\left(\frac{Y_B^+}{30}\right)\right]\right\} \quad \text{for} \quad Y_B^+ > 30 \tag{67}$$

The dimensionless distance Y_B^+ can be estimated by [38]:

$$Y_B^+ = 0.015\sqrt{\sigma_f d_{in}\rho_f}/\mu_f \tag{68}$$

In the present study, the B-coefficient method [39] was employed to calculate the friction coefficient. In that, the two-phase frictional multiplier can be given from:

$$\phi_{LO}^2 = \Delta p_{tp}/\Delta p_{lo} \tag{69}$$

where the two-phase frictional resistance coefficient can be given from:

$$f\left(x\right) = \frac{\phi_{LO}^2 - 1}{\left(\rho_l/\rho_g - 1\right)} \tag{70}$$

Rewrite Eq. (70) as

$$f\left(x\right) = Bx\left(1-x\right) + x^2 \tag{71}$$

Considering that B is determined by the density ratio of vapor and liquid, the following form was assumed:

$$f\left(x\right) = B_0 x^n\left(1-x\right)^m + x^2 \tag{72}$$

From regression of the test data, we obtained:

$$f\left(x\right) = 1.426\times x^{1.85}\times\left(1-x\right)^{0.532} + x^2 \tag{73}$$

Finally, the bubble detachment diameter [40] can be determined as:

$$D_b = 0.04\frac{c_{pf}\rho_f T_{sat}}{\rho_g h_{fg}}\sqrt{\frac{2\sigma}{g\left(\rho_f-\rho_g\right)}} \tag{74}$$

4.2.5 Drag force exerted on a bubble

The F_d can be given from:

$$F_d = \frac{\lambda\rho_c\overline{U}_c^2}{2}\left(\frac{\pi D_b^2}{4}\right) \tag{75}$$

The λ [38] can be given from:

$$\frac{1}{\sqrt{\lambda}} = 3.48 - 4\lg\left(\frac{2D_b}{d_{in}} + \frac{9.35}{Re_{2\phi}\sqrt{\lambda}}\right) \tag{76}$$

where

$$Re_{2\phi} = Gd_{in}/\mu_{2\phi} \tag{77}$$

The average viscosity of main flow can be calculated from the formula of Beattie and Whalley [41]:

$$\mu_{2\phi} = \mu_l\left(1-\alpha_{tr}\right)\left(1+2.5\alpha_{tr}\right) + \mu_g\alpha_{tr} \tag{78}$$

4.2.6 Average flow velocities

Since the bubble layer is very thin, the velocity distribution is assumed to be linear. Thus, the average flow velocity in the bubble layer is given as:

$$\overline{U}_b = 0.5U_c \tag{79}$$

The velocity of the main flow can be calculated as:

$$U^+ = \begin{cases} Y^+ & Y^+ < 5 \\ 5\ln Y^+ - 3.05 & 5 \le Y^+ < 30 \\ 2.5\ln Y^+ + 5.5 & Y^+ \ge 30 \end{cases} \tag{80}$$

where

$$U^+ = U_c\sqrt{\frac{\rho_c}{\tau_w}} \tag{81}$$

$$Y^+ = \frac{D_b\sqrt{\tau_w\rho_c}}{\mu_{2\phi}} \tag{82}$$

$$\tau_w = \frac{\lambda G^2}{2\rho_c\eta_c^2} \tag{83}$$

Thus, the average flow velocity in the bubble layer can be determined from Eq. (79) given Eq. (80). According to the velocity profile in Eq. (80), the main flow region average velocity can be given as:

$$\overline{U}_c = \frac{G - \overline{U}_b\rho_b(1-\eta_c)}{\rho_c\eta_c} \tag{84}$$

4.2.7 Other relations

We believe that a bubbles layer can block heat transfer and trigger CHF. Thus, the bubble-layer thickness is assumed as:

$$\delta_b = D_b \tag{85}$$

4.3 Calculation procedure, comparison of constitutive models, and validation

4.3.1 Calculation procedure

To find q_{cr} from Eq. (49), the following calculation procedure can be employed, as described in Fig. 13.

4.3.2 Validation

As depicted in Fig. 14 and Fig. 15, the prediction by the established CHF model is compared with experimental data attainable in Refs. [42, 43]. Table 3 shows the experimental data used to validate the model. The relative errors of the accuracy of the Thompson, Becker and experimental data with the newly established model are 20.5%, 19.5% and 5.7%. The predicted CHF is lower than the experimental CHF as the true vapor quality exceeds 0.5. It indicates that this model is inapplicable in that region. In Fig. 15, we find that the model's accuracy deteriorates as the mass flux is larger than 800 kg·m^{-2}·s^{-1}, which may be because the model adopted for the frictional resistance coefficient is no longer applicable. Fig. 16 shows the comparison of the test result, empirical formula and model calculation. We can see from Fig. 16, the multivariate GPR method, deep neural network method and the calculation model show good calculation results.

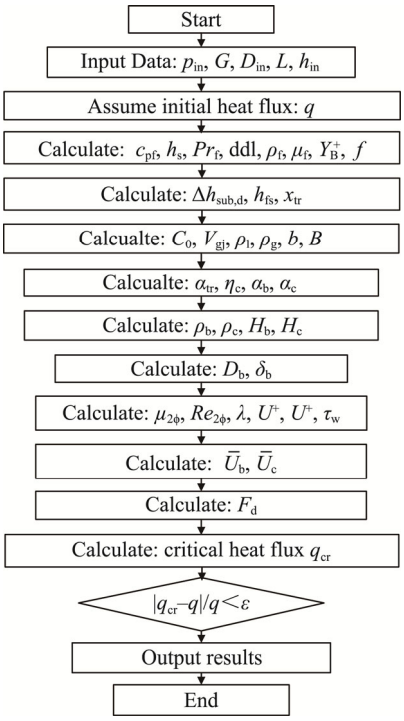

Fig. 13 Flow diagram of the calculation procedure for the proposed CHF model

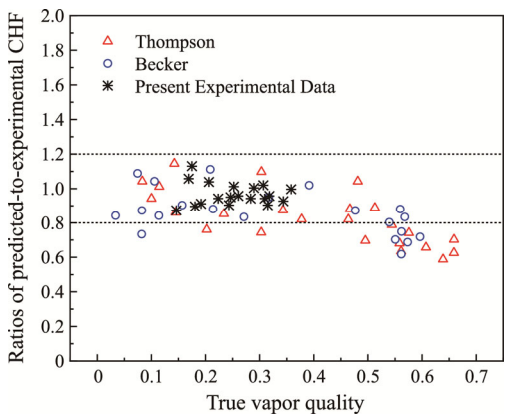

Fig. 14 Ratios of the calculated-to-experimental CHF vs. the true vapor quality

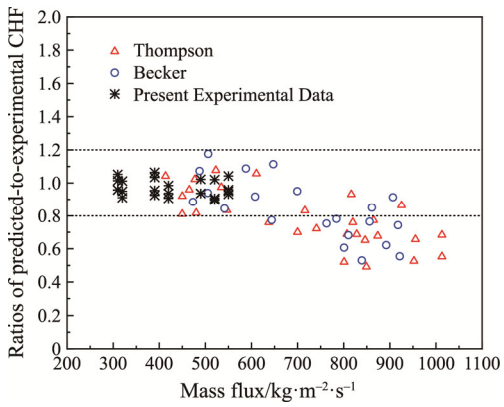

Fig. 15 Ratios of the calculated-to-experimental CHF vs. mass flux

Table 3 Experimental data used to validate the model

Reference	p /MPa	G /kg·m^{-2}·s^{-1}	ΔH_{in} /kJ·kg^{-1}	q /kW·m^{-2}
Thompson [42]	13–19	400–1000	10–260	400–1000
Becker [43]	17–20	452–980	60–300	200–800
Present Experimental Data	18–21	310–550	30–100	200–470

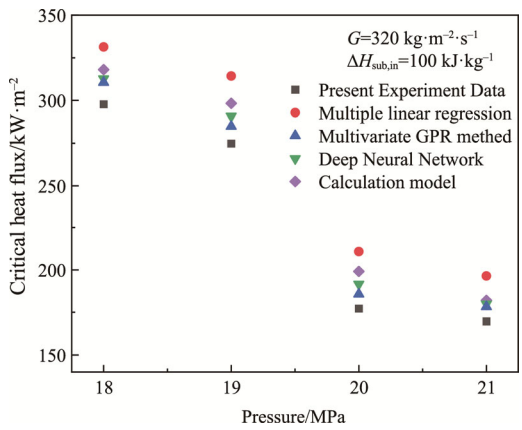

Fig. 16 The comparison of the experimental result, empirical formula and model analysis

4.4 Model predictions and analysis

We analyze the predictions of the established theoretical model at varying pressure, mass flux, inlet subcooled enthalpy, tube length, and tube diameter. The model predictions focused on the subcooled boiling zone and the saturated boiling zone with low vapor quality.

4.4.1 Effect of pressure on CHF

The true vapor quality vs. CHF at four near-critical pressures within 18–21 MPa is examined in Fig. 17. The CHF is found decreasing with the increase of vapor quality, which can be due to that the bubble nucleation is incrementally developed. Fig. 17 shows when the true vapor quality is 0.2, the values of the CHF corresponding to four near-critical pressures are 286, 369, 452 and 521 kW·m^{-2}, respectively. The CHF is smaller when the pressure is closer to 22.115 MPa. These findings agree with the experimental results.

4.4.2 Effect of mass flux on CHF

Fig. 18 shows the variation of CHF when the mass fluxes increase from 350 to 800 kg·m^{-2}·s^{-1}. Fig.18 shows the values of the CHF corresponding to four mass fluxes are 198, 249, 301 and 352 kW·m^{-2} when the true vapor quality is equal to 0.2. The CHF is found smaller for smaller mass flux due to the fact that a larger mass flux can take away bubbles near the wall, and as a result more water can exchange heat with the wall. This finding agrees with the experimental results.

4.4.3 Effect of inlet subcooled enthalpy on CHF

We predict the variation of CHF with the increase of ΔH_{in}. The different ΔH_{in} means different inlet subcooled temperature. The results are presented in Fig. 19. The CHF increases with the increase of ΔH_{in}. The reason may be that the fluid needs more heat flux to complete the process of bubble nucleation. With fewer bubbles formed on the wall, it is more difficult for heat transfer deterioration to occur.

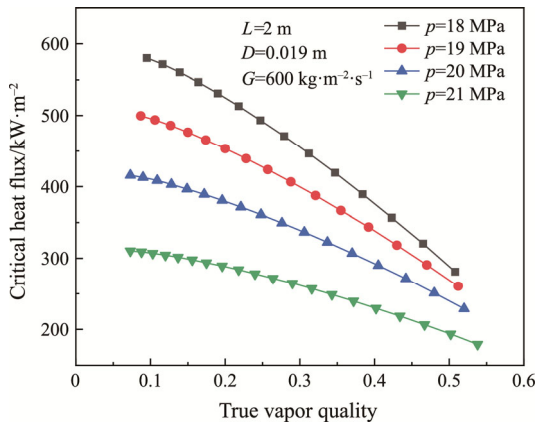

Fig. 17 Effect of pressure on calculated q_{cr}

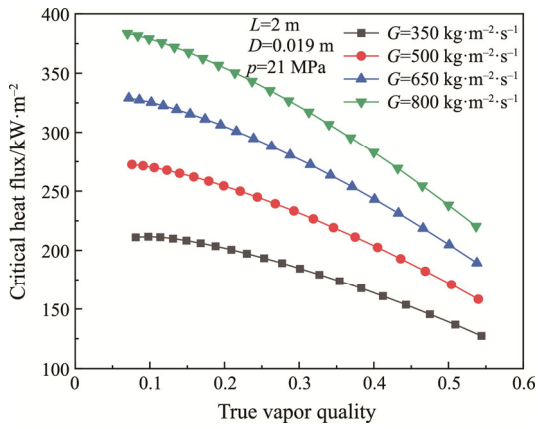

Fig. 18 Effect of mass flux on calculated q_{cr}

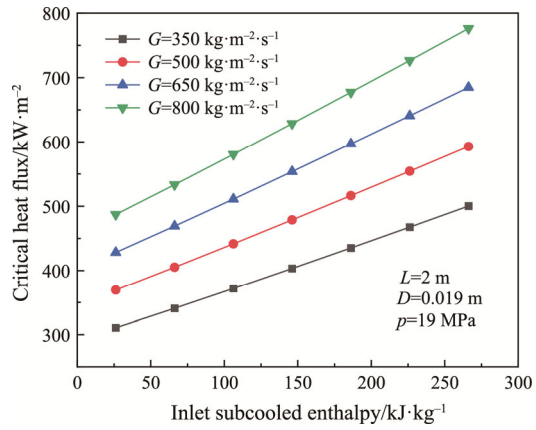

Fig. 19 Effect of inlet subcooled enthalpy on calculated q_{cr}

4.4.4 Effect of tube length on CHF

We applied the CHF model to predict the variation of CHF with increasing length of the heated tube at three different mass fluxes. Fig. 20 shows that the CHF will decrease if heated tube's length is increased. The reason may be that the fluid can gain more heat in the longer tube, making heat transfer deterioration easier to occur.

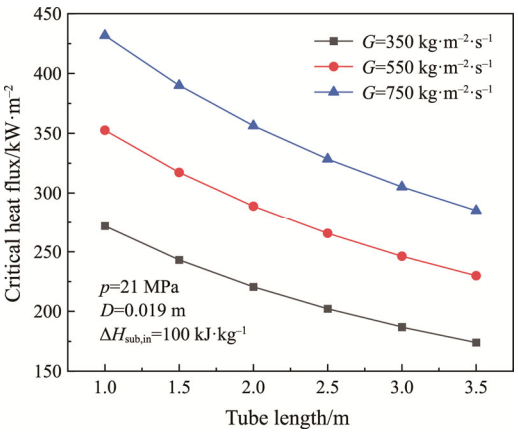

Fig. 20 Effect of tube length on calculated q_{cr}

4.4.5 Effect of tube diameter on CHF

The variation of CHF with the increase of tube diameter was analyzed in Fig. 21. If the diameter is decreased, the CHF will decrease, resulting from the fact that the fluid needs more heat flux in a wider tube to complete the process of bubble nucleation. With fewer bubbles formed on the wall, it is more difficult for heat transfer deterioration to occur.

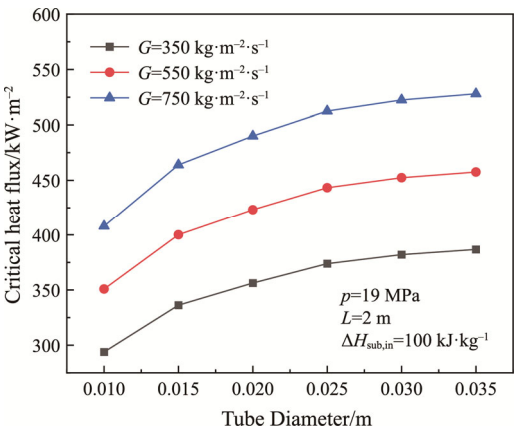

Fig. 21 Effect of tube diameter on calculated q_{cr}

5. Conclusions

We have presented an experimental and theoretical study on the CHF in a vertical up smooth water wall tube of the 660 MW efficient USCFB boiler under near-critical pressures. Our main findings are summarized as follows.

(1) The near critical pressure CHF test data of water are obtained from our experimental study. We have found heat transfer deterioration of USCFB boiler water wall tube (12Cr1MoVg) mainly occurs when the vapor quality is less than 0.4 and is the DNB-type. The CHF decreases when the pressure increases. It is because the density difference between saturated steam and water decreases when the pressure increases, which reduces the buoyancy and makes it more difficult for bubbles to escape from the wall. The CHF decreases with the decreases of the mass flux. It is because a larger mass flux can take away bubbles near the wall, and as a result, more working medium can exchange heat with the wall.

(2) A correlation function for the CHF considering the key parameters: pressure, mass flux, and inlet subcooling temperature, has been inferred from the experimental data. To this end, the machine/deep learning methods such as the multivariate GPR and DNN outperform the traditional regression method such as the multiple linear regression due to the complex nonlinear relationship between the CHF and key parameters. Correlations established via machine learning greatly improved the regression accuracy.

(3) In our theoretical study, an improved CHF model has been established, which is based on the near-surface bubble crowding model. The agreement between the predictions on the CHF by the established theoretical model and different experimental results reported in literature validates our CHF model and also demonstrates that the experimentally observed CHF is indeed the DNB-type.

(4) With the established theoretical CHF model, we have predicted the dependence of the CHF on different quantities including the true vapor quality, inlet subcooled enthalpy, and length and diameter of the heated tube. Thus, it can provide reference for the future optimal design of the tube in the USCFB boiler.

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References

[1] Koornneef J., Junginger M., Faaji A., Development of fluidized bed combustion—An overview of trends, performance and cost. Progress in Energy and Combustion Science, 2007, 33: 19–55.

[2] Lü J.F., Zhang M., Yang H.R., Liu Q., Liu Z.Q., Zhao Y.G., Conceptual design of a simplified 660 MW

ultra-supercritical circulating fluidized bed boiler. Proceedings of the CSEE, 2014, 34(5): 741–747. (in Chinese)

[3] Ling W., Lyu J.F., Zhou T., Yang H.R., Chen Y., Li R.H., Research and development progress of the 660 MW ultra-supercritical circulating fluidized bed boiler. Proceedings of the CSEE, 2019, 39(9): 2515–2523. (in Chinese)

[4] Wang W.Y., Yang D., Jiang H.Q., Qu M.F., Zhao Y.J., Heat transfer and frictional resistance characteristics of the water wall tube of an ultra-supercritical CFB boiler. The Journal of Supercritical Fluids, 2017, 128: 279–290.

[5] Wang W.Y., Yang D., Liang Z.Y., Qu M.F., Ouyang S.J., Experimental investigation on flow instabilities of ultra-supercritical water in parallel channels. Applied Thermal Engineering, 2019, 147: 819–828.

[6] Wang W.Y., Yang D., Dong L., Zhou X.H., Li J., Pan W.X., Experimental and numerical study on density wave oscillations of supercritical water in parallel water wall channels of an ultra-supercritical circulating fluidized bed boiler. Applied Thermal Engineering, 2020, 165: 114584.

[7] Kossolapov A., Chavagnat F., Nop R., Dorville N., Phillips B., Buongiorno J., Bucci M., The boiling crisis of water under exponentially escalating heat inputs in subcooled flow boiling at atmospheric pressure. International Journal of Heat and Mass Transfer, 2020, 160: 120137.

[8] Liu Q.Q., Kelly J., Sun X.D., Flow regime transition in the post-CHF flow regimes under subcooled and low-quality conditions. International Journal of Multiphase Flow, 2021, 136: 103543.

[9] Liu B., Yuan B., Zhou J., Zhao J.F., Marco P.D., Zhang Y.H., Wei J.J., Yang Y., Analysis of the critical heat flux of subcooled flow boiling in microgravity. Experimental Thermal and Fluid Science, 2021, 120(1): 110238.

[10] Tan L.Z., Chen C.N., Dong X.M., Gong Z.Q., Wang M.T., Experimental study on CHF of R134a flow boiling in a horizontal helically-coiled tube near the critical pressure. Experimental Thermal and Fluid Science, 2017, 82: 472–481.

[11] Mastrullo R., Mauro A.W., Viscito L., Experimental CHF for low-GWP fluids and R134a. Effect of the L_h/D ratio at low and high mass velocities. International Journal of Heat and Mass Transfer, 2017, 109: 1200–1216.

[12] Dang C., Jia L., Peng Q., Huang Q., Zhang X., Experimental and analytical study on nucleate pool boiling heat transfer of R134a/R245fa zeotropic mixtures. International Journal of Heat and Mass Transfer, 2018, 119: 508–522.

[13] Li X.J., Jia L., Dang C., An Z.J., Huang Q., Visualization of R134a flow boiling in micro-channels to establish a novel bubbly-slug flow transition criterion. Experimental Thermal and Fluid Science, 2018, 91: 230–244.

[14] Mouslim A., Tavoularis S., Direct tests of fluid-to-fluid scaling expressions for supercritical heat transfer in tubes. International Journal of Heat and Mass Transfer, 2020, 147: 118862.

[15] Zahlan H., Groeneveld D.C., Tavoularis S., Fluid-to-fluid scaling for convective heat transfer in tubes at supercritical and high subcritical pressures. International Journal of Heat and Mass Transfer, 2014, 73: 274–283.

[16] Zhao X., Shirvan K., Salko R.K., Guo F., On the prediction of critical heat flux using a physics-informed machine learning-aided framework. Applied Thermal Engineering, 2020, 164: 114540.

[17] Hobold G.M., Silva A.K., Visualization-based nucleate boiling heat flux quantification using machine learning. International Journal of Heat and Mass Transfer, 2019, 134: 511–520.

[18] Kam D.H., Jeong Y.H., NO H.C., A heat transfer model development for CHF prediction with consideration of dry patch characteristics. International Journal of Heat and Mass Transfer, 2020, 148: 118–128.

[19] Zhang H., Mudawar I., Hasan M.M., Experimental and theoretical study of orientation effects on flow boiling CHF. International Journal of Heat and Mass Transfer, 2020, 45: 4463–4477.

[20] Weisman J., Pei B.S., Prediction of critical heat flux in flow boiling at low qualities. International Journal of Heat and Mass Transfer, 1983, 26(10): 1463–1477.

[21] Lee C.H., Mudawwar I., A mechanistic critical heat flux model for subcooled flow boiling based on local bulk flow conditions. International Journal of Multiphase Flow, 1988, 14(6): 711–728.

[22] Chang S.H., Lee K.W., A critical heat flux model based on mass, energy, and momentum balance for upflow boiling at low qualities. Nuclear Engineering and Design, 1989, 113(1): 35–50.

[23] Kwon Y.M., Chang S.H., A mechanistic critical heat flux model for wide range of subcooled and low quality flow boiling. Nuclear Engineering and Design, 1999, 188: 27–47.

[24] Wang W.Y., Yang D., Liu W.Y., Nie X., Wang S.Y., Bi Q.C., Experimental investigation on two-phase flow instabilities of ultra supercritical pressure boiler with spirally water wall tubes. Experimental Thermal and Fluid Science, 2017, 81: 304–313.

[25] Shen Z., Yang D., Xie H.Y., Nie X., Liu W.Y., Wang S.Y., Flow and heat transfer characteristics of high-pressure water flowing in a vertical upward smooth tube at low mass flux conditions. Applied Thermal Engineering, 2016, 102 (5): 391–401.

[26] Moffat R.J., Describing the uncertainties in experimental results. Experimental Thermal and Fluid Science, 1988, 1: 3–17.

[27] Montgomery D.C., Peck E.A., Vining G.G., Introduction to linear regression analysis, fifth ed., MIT Press, Cambridge, 2012.

[28] Rasmussen C.E., Williams C.K.I., Gaussian processes for machine learning, fourth ed., MIT Press, Cambridge, 2006, pp. 79–104.

[29] Jürgen S., Deep learning in neural networks: An overview. Neural Networks, 2015, 61: 85–117.

[30] Bengio Y., Learning deep architectures for AI. Foundations and Trends® in Machine Learning, 2009, 2(1): 1–127.

[31] Origin. OriginLab Corporation, Northampton, USA, 2018.

[32] Xie H.Y., Yang D., Wang W.Y., Wan L., Li C.M., An improved DNB-type critical heat flux prediction model for flow boiling at subcritical pressures in vertical rifled tube. International Journal of Heat and Mass Transfer, 2019, 132: 209–220.

[33] Kwon Y., Chang S., A mechanistic critical heat flux model for wide range of subcooled and low quality flow boiling. Nuclear Engineering and Design, 1999, 118: 27–47.

[34] Zuber N., Findlay J.A., Average volumetric concentration in two-phase flow systems. International Journal of Heat and Mass Transfer, 1965, 87(4): 453–468.

[35] Lahey R., Moody F.J, The thermal-hydraulics of a boiling water nuclear reactor, second ed., American Nuclear Society, United States, 1993.

[36] Saha P., Zuber N., Point of net vapor generation and vapor void fraction in subcooled boiling. 5th Int. Heat Transfer Conference, Tokyo, Japan, 1974, 1: 175–179. DOI: 10.1039/AN9871200259.

[37] Levy S., Forced convection subcooled boiling: Prediction of vapor volumetric fraction. International Journal of Heat and Mass Transfer, 1967, 10(7): 951–965.

[38] Colebrook C., White C., Experiments with fluid friction in roughened pipes. Proceedings of the Royal Society of London, 1937, 161(906): 367–381.

[39] Chisholm D., Two-phase flow in pipelines and heat exchangers, first ed., The Institution of Chemical Engineers, London, 1983.

[40] Cole R., Rohsenow W.M., Correlation of bubble departure diameters for boiling of saturated liquids. Chemical Engineering Progress Symposium Series, 1969, 65(92): 211–213.

[41] Beattie D., Whalley P., A simple two-phase frictional pressure drop calculation method. International Journal of Multiphase Flow, 1982, 8(1): 83–87.

[42] Thompson B., Macbeth R.V., Boiling water heat transfer in uniformly heated round tubes: A compilation of world data with accurate correlations. AEEW-R-356, United Kingdom Atomic Energy Authority, 1964.

[43] Becker K.M., Djursing D., Lindberg K., et al., Burnout conditions for round tubes at elevated pressures. Proceedings of the International Symposium on Two-Phase Systems, 1972, pp. 55–73.